FYP SRS Document

Final Year Project

Software Requirement Specification

For

Try & Buy

BSCS

By

S#	Name	Registration #/Roll	Mobile #	E-Mail
1.	Anum Shahzadi	Fa-2022/BSCS-459	0335-4499185	Fa22-BSCS-459@cs.lgu.edu.pk
2.	Misbah Ayaz	Fa-2022/BSCS-100	0326-5043101	Fa22-BSCS-100@cs.lgu.edu.pk

Supervised by:

Sir Khurram Yussouf Tehseen _____ **(**Signature**)**

CS Department
Lahore Garrison University
Lahore

Table of Contents

1.1 Purpose 2

1.2 Document Conventions 2

This document is for: 2

1.4 Product Scope 2

2. Overall Description 3

2.1 Product Perspective 3

2.2 Product Functions 3

2.3 User Classes and Characteristics 3

2.4 Operating Environment 4

2.5 Design and Implementation Constraints 5

2.6 User Documentation 5

2.7 Assumptions and Dependencies 5

3. External Interface Requirements 7

3.1 User Interfaces 7

3.2 Hardware Interfaces 7

3.3 Software Interfaces 8

3.4 Communications Interfaces 8

4. System Features 9

4.1 Picture Upload Feature 9

4.1.1 Description and Priority 9

4.1.2 Stimulus and Response series 9

4.1.3 Functional Requirements 10

4.2 Clothing Selection Feature 10

4.2.1 Description and Priority 10

4.2.2 Stimulus/Response Sequences 10
4.2.3 Functional Requirements 11
4.3 Virtual Try-On Generation 11
4.3.1 Description and Priority 11
4.3.2 Stimulus/Response Sequences 11
4.3.3 Functional Requirements 12
4.4 Admin Panel Feature 12
4.4.1 Description and Priority 12
4.4.2 Stimulus/Response Sequences 12
4.4.3 Functional Requirements 13
5. Nonfunctional Requirements 13
5.1 Performance Requirements 13
5.2 Safety Requirements 13
5.3 Security Requirements 14
5.4 Quality Attributes of software 14
5.5 Business Rules 15
5.6 Use Case Diagram 17
5.7 Architecture Diagram 18
5.8 Activity Diagram 19
5.9 Sequence Diagram 20
5.10 ER Diagram 21
References 22

1. Introduction

1.1 Purpose

The proposed Software Requirements Specification document is intended to specify both functional requirements and non-functional requirements of Try and Buy an AI based virtual try on clothing system[1]. The system enables user to leave a photo, select piece of clothing and see how he/she would look in the clothing. The SRS concerns the whole system with back end processing with MediaPipe [2], [10], [11], openCV [3], [15], GAN-based models [12] as well as front end user interface.

1.2 Document Conventions

The subheadings are arranged in a hierarchical fashion (1, 1.1, 1.1.1). The code REQ-# represents the functional requirements. The technical terms (such as GAN, OpenCV, AR) are typed in upper case. Any diagrams and photographs will be in line with the principles of university formatting [4],[5]

1.3 Suggestions of the intended audience and reading

This document is for:

Developers: to know the components and the technical requirement.

Supervisors: to check the scope and definition of the project.

Testers: to know about what should be expected and what should be tested.

Project Members: to consult in the development.

These are to start at Introduction, and move on to Overview Description, and then as System features and then Non- Functional Requirements [4],[5].

1.4 Product Scope

Try & Buy is dedicated to improving the online clothing retailing enterprise by providing individuals with the chance of seeing the clothes on their bodies through the usage of Artificial Intelligence. It is a system that is used to overlay a few clothes to the picture of the user using image processing and deep learning models. It will play a role in instilling confidence with the consumers, in reducing the rate of returns and offering more of interactive shopping experience to the retailers[1].

2. Overall Description

2.1 Product Perspective

Try & Buy is a new web-based application that we created on the basis of the intention to increase the experience of online shopping of clothes and virtual try-on with the help of the application of artificial intelligence[1]. Unlike traditional e-commerce systems that are based on static images and size charts, this system enables users to see clothing on their own images. The product can be used as a standalone application or integrated with existing e-commerce platforms using the use of the API in the form of a "RESTful" application. With an integrated approach, user accounts and payments are handled by an external system, whereas Try and Buy is concentrated on the virtual try-on feature. The system has three major parts: a web-based user interface, a back-end application server, and an AI processing module based on MediaPipe [2],[10],[11], OpenCV [3] and GAN-based models [12]. These modules are communicating via secure APIs that guarantee a maintainable, scalable and modular architecture [4],[5].

2.2 Product Functions

The following are the main functions offered by the Try & Buy system:

i. Enable users to post or take their pictures in order to try virtually.
ii. Check uploaded images in terms of format, clarity and body visibility.
iii. Present an online fashion catalog to browse through and pick.
iv. Make the user be able to select and alter clothes prior to processing.
v. Recognize body poses with AI-based models [2],[10],[11].
vi. Create life-like virtual try-on performances by superimposing chosen clothes [12].
vii. Present the produced try-on image to the customer.
viii. Give the user the option to redo or change the clothes
ix. Has an administrative panel to administer clothing catalog and system data.

All these functions exhibit the virtual try-on process and improve the online shopping process [1].

2.3 User Classes and Characteristics

General Users (Customers):

Primary users of the system, who upload images, select the clothing and see virtual try-on results. One/2/6 One or two people or the Six People They have basic technical knowledge & Use occasionally or frequently

Registered Users:

Users with accounts who are able to save try on results, view history, manage preferences and make purchases. They are interacting more frequently with the system.

Administrators:

Authorized users in charge of managing the clothing catalog and data of the system. They have higher technical expertise and limited access privileges.

General and registered users are the most important classes of users, and administrators are used to maintain and control the system.

2.4 Operating Environment

Try and Buy system will operate in web-based environment and the following configuration will be interested:

Client Side:

i. Devices: Tablets, computers, laptops and phones.

ii. OS: Windows 10+, Android

iii. Browsers: Google Chrome

Server Side:

iv. python Backend Framework: Python Flask [14].

v. AI Libraries MediaPipe [2], [10], [11] , OpenCV [3], TensorFlow/PyTorch [13].

vi. Database: MongoDB [7]

Network Environment:

vii. Uninterrupted internet connectivity to upload pictures and do AI processing.

viii. Supports mobile (4G/5G) network and broadband.

ix. The system will be deployed in a way that it does not affect the normal web services, databases, and third-party APIs.

2.5 Design and Implementation Constraints

The design and implementation has the following limitations. To implement the pose detector and the virtual try-on processing, the system must make use of MediaPipe [2], [10], [11], OpenCV [3] and GAN-based models [12]. The availability of GPU resources, memory capacity and server processing power is a limitation to AI performance. The uploaded images should be limited in format and size, so as to obtain proper processing of the image.The system is restricted by the web-based architecture, where direct access to the native device capabilities cannot be made. Safe communication should be adopted using the assistance of https and restful APIs. The processing of user data must comply with data privacy and security policies. The system is created under the time limitation of scholars and minimal training data sets. It should be implemented as the standard coding practices and documentation guidelines to be implemented to offer maintainability [4], [5]. Such limitations influence architectural design, system choice and performance.

2.6 User Documentation

The Try and Buy system will be accompanied with the following user documentation:

User Manual: An online tutorial on how to upload images, pick the clothes, and have an online try-on.

Admin Manual: Documentation to the administrators which is a description of the system controls and how the catalog should be managed.

Online help: Help and tooltips ,Context-based help and tooltips are components of the application interface.

Tutorial Screens: Guides to first time users.

All the documents are going to be delivered in the electronic form (PDF and web-based sources) and simple and easy to use language with the norm of standard rules of documentation [4], [5].

2.7 Assumptions and Dependencies

The following assumptions have been made concerning the model:

The Try & Buy system is being developed based on the following assumptions: The picture will be uploaded by the user in the situation when the victim has adequate illumination and the upper part of the body is not covered. Internet connectivity will be made available to users. Artificial intelligence will provide a reasonable accuracy in the case of common body postures and clothing type [2], [10], [11], [12]. Modern web standards will be used as web browsers that the users will be using. This system is dependent on the following: Access to and continued maintenance of third party libraries like MediaPipe [2], [10], [11], OpenCV [3], TensorFlow/PyTorch [13]. Availability of sufficient GPU and servers to perform running of the AI model. Relying on the web frameworks (React.js [8]) to execute the functionality both at the frontend and backend Integration with Database services (MongoDB [7]) to store the data. Reliance on third-party payment gateways (non-obligatory, along with an e-commerce system). Any alterations in these assumptions or dependencies can affect the performance of the system, accuracy, or its availability.

3. External Interface Requirements

3.1 User Interfaces

Try and Buy application consists of a well built web interface that is clean and modern. The interface has implemented the ethos of responsive design so as to adopt the same functionality and appearance on the desktops, tablets as well as mobile devices. The key components of the interface include Upload Feature: The button to upload an image is very prominent, the user can either choose an image file on the phone or directly use the camera in the phone to make a snap. This component has drag and drop. Clothing Selection Gallery: Clothing Selection Gallery is a gallery of clothes with plenty of pictures, both in the form of a carousel. virtual try on preview area: This is a special canvas, where the live virtual try on result can be seen, in this section, users can view the clothes on their uploaded photos. Error Handling: In other instances like unwanted image formats, poor lighting, bad pose, and others, a modal pop-up will issue easy to comprehend error messages to the user. User Account Dashboard: Registered users also receive their own dashboard where they can view their history of trying on items, their favorite items since they have already liked them, their size preferences as well as the past purchases so that they can have a quick reference and can also replenish them.

3.2 Hardware Interfaces

The system will be interrelated with some hardware parts to give it its primary functionality: Garners optional access to the user device cameras via standardized web applications, Real-time images capturing that can be utilized to hold an instant try-on session. The system is aware of the presence of cameras and tunes the settings of the interface. Server Processing Hardware Special hardware configurations Server hardware optimized to execute AI / ML workloads. It has of high-performance processing units of the graphics processing unit (GPUs) (compatible with Nvidia CUDA technology) to execute the inference process of the neural networks more quickly than any alternative hardware, high-enough RAM to load the model and to execute data simultaneously and high-speed storage devices to handle the information easily. Display Hardware Compatibility: Compatible with several display modes including touch-screen display mode (mobile devices) and normal desktop display so that the same visual quality is maintained across all display modes. Network Infrastructure: Internet connectivity at the client and server office is good and bandwidth is not high since this is to enable transfer of images and real time processing feedback.

3.3 Software Interfaces

The software can be defined as a combination of different software components in the program with clearly defined software interfaces: Front-End Framework: React. Js architecture [8] as primary architecture of the component-based architecture. The interface is built based on responsive CSS frameworks (such as Tailwind CSS) to ensure the style of the device is identical. Back-End Services: python (Flask) server architecture [14] to implement the business logic and the API interfaces and connection to AI modules. The back-end is developed based on the notion of the Restful design and contains sufficient error management and validation of the requests. In a real time body pose estimation and landmark detection, the skeletal data is required to perfectly position a garment. [2], [10], [11] MediaPipe Integration [2],[10],[11] In a real time body pose estimation and landmark detection, the skeletal data is required to accurately position a garment. OpenCV Processing [3]: To carry out certain tasks associated with image pre-processing like resizing, normalization, elimination of the background and also the color correction. Generative Adversarial Networks (GANs) [12]: Longer Models of Photorealistic Garment Synthesis and Garment Overlay Trained on Fashion Databases Management: MongoDB [7] To organize the information of the database, such as users, different products catalogs, order history and transaction, etc. with the necessary indexing to be efficient. Third-Party Integration - Payments (Digital-Banking/ PayPak).

3.4 Communications Interfaces

The system employs secure system communications between components: Encrypted Data transfer: With HTTPS, 1.2+ built on TLS to ensure that all clientserver data is encrypted to make sure that sensitive user data is not lost in transit: images and personal information. Other supplementary security measures. RESTful API Architecture: Reasonable layering of front-end and back-end based on REST APIs based on the traditional conventions of the HTTP methods, status code and naming of resources. Data Exchange Format: Data exchange format (JSON) is used to represent all the requests and responses that the API makes. Real-time Communication: WebSocket connection to be updated on the live updates as the AI is being processed and the user is displaying his/her progress.

4. System Features

4.1 Picture Upload Feature

4.1.1 Description and Priority

This is the fact that its users can add their frontal photo to commence with the virtual try-on experience. The system verifies the image format and the visibility of upper body and then processes.

Priority: High

Component	Rating (1–9)
Benefit	9
Penalty (if missing)	8
Cost	3
Risk	4

4.1.2 Stimulus and Response series

Stimulus (User Action)	Response (System Behavior)
User clicks on Upload Image button	System opens file selection dialog.
User selects JPG/PNG picture	System uploads and scans the image.
Image is posted by the user System	checks the image with respect to the format and clarity.
In case of validity	Moves to next step (clothing selection)
In case of invalid	Error message on the system (Image too blurry or incomplete)

4.1.3 Functional Requirements

REQ-1 System must be able to upload .jpg and .png files.

REQ-2 System shall ensure that there is a human in the picture (face and some upper body).

REQ-3 System should inform the user in case of uploaded image being either too blurred or incomplete.

REQ-4 System should not permanently store images against user choice.

REQ-5 System shall remove temporary images once the processing is done.

4.2 Clothing Selection Feature

4.2.1 Description and Priority

The feature enables users to search, sort and select clothing in the catalog. The chosen product of clothing is uploaded to the AI engine where it is also tried on.

Priority : High

Component	Rating (1–9)
Benefit	8
Penalty	7
Cost	2
Risk	3

4.2.2 Stimulus/Response Sequences

Stimulus (User Action)	Response (System Behavior)
Click on catalog	Items loaded to grid view.
User clicks any clothing System	Presents information.
User clicks on try on System	forwards item + image to processing.
In case of failure	to retrieve item in the system Displays item unavailable message.

4.2.3 Functional Requirements

REQ-6 System should display different items of clothing in catalog.

REQ-7 System must get information about items (size, type, image).

REQ-8 System must be able to accommodate a change of clothing option before the visualization of the results.

REQ-9 System must show an error in case of the unavailability of clothing data.

4.3 Virtual Try-On Generation

4.3.1 Description and Priority

By means of this feature, the AI models (MediaPipe, OpenCV, GAN) are used to overlay the chosen clothing on the image of a user in their real image and present the final composite image.

Priority: Very High

Component	Rating (1–9)
Benefit	10
Penalty	9
Cost	7
Risk	6

4.3.2 Stimulus/Response Sequences

Stimulus (User Action)	Response (System Behavior)
Image + clothing item are inputted into the system	The AI engine begins detecting body keypoint.
Keypoints found	Clothing matching process starts.
Alignment done	GAN results in realistic overlay.
Finalizing completion	Try on image shown to user.
In case of processing failure	Display “Could not give a result, please restart over again"

4.3.3 Functional Requirements

REQ-10 System should be able to identify body keypoints via MediaPipe [2], [10], [11].

REQ-11 System must position clothing on user according to the identified keypoints.

REQ-12 System should generate realistic try-on result with the use of GAN [12].

REQ-13 System must display output image on successful processing.

REQ-14 System should display amiable re-try message in case of processing failure.

4.4 Admin Panel Feature

4.4.1 Description and Priority

Admin Panel allows authorized users with the role of an administrator to manage clothing catalog, control the work of the system, and guarantee the accuracy of data.

Priority: Medium

Component	Rating (1–9)
Benefit	7
Penalty	5
Cost	4
Risk	5

4.4.2 Stimulus/Response Sequences

Stimulus (User Action)	Response (System Behavior)
Admin enters login credentials	System verifies authentication
Admin selects “Add Clothing”	System opens upload form
Admin uploads clothing item	System stores item in catalog database
Admin deletes or updates item	Catalog updates instantly on frontend
If unauthorized access attempt	System denies access (error message)

4.4.3 Functional Requirements

REQ-15 System must forbid access to panel without authentication of the user (admin).

REQ-16 Admin should have access to uploading new clothes with images.

REQ-17 Admin should have the capability of updating/editing clothing data.

REQ-18 Admin should be capable of removing pieces of clothing.

REQ-19 System should maintain logs of the activities of the administrators to ensure security.

REQ-20 Catalog modifications can only be accessed by the admin.

5. Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time**

The loading time of the page will be less than 3 seconds on 4G connection. Standard image processing will occupy 5 seconds of try on. The findings will be presented within under 3-4 seconds.

- **Throughput**

System will support 100 users at the peak times. The AI processing pipeline will have the capability to handle 20 simultaneous try-ons. There will be 1000 read operations per second in database.

- **Scalability**

System is going to scale horizontally up to the level of 10x increment of traffic volume. The deployment of intelligent modules will take the form of micro services. Database will be supporting performance replicas.

5.2 Safety Requirements

- **Content Safety & Moderation**

The system will avoid downloading of indecent or malicious contents.

- **Security and Compliance of payments**

The issue of receiving user payments will be performed using certified and secure payment gateways.

- **Data Integrity System Resilience**

System will not allow the loss of data when the server suddenly cuts, automatic saving of data will be done periodically in checkout and authentication process.

5.3 Security Requirements

- **Data Protection**

TLS 1.3 will be used to encrypt any information that is sent by the user. Passwords hashing will be carried out. Encryption of personal images when at rest will be applied. System will be in compliance with GDPR data protection.

- **Access Control**

Role based access control on administrators and customers. The session tokens will be of 24 hour expiry. Upon 5 attempts to log-in with invalidity, account lockout will be enabled.

5.4 Quality Attributes of software

- **Usability**

The UI will be friendly enough even to simple digital literate users. The users will have an opportunity to use the whole process of try-on, 3-5 times. The instructions on how to upload the images will be in the visual form and therefore reduce the risk of the user making mistakes.

- **Reliability**

In normal conditions, the system would generate the correct functional output on 95 percent of the try-on requests. The automated error handling will also be incorporated in the system to prevent sudden crash of the application.

- **Maintainability**

A modular design will be utilized in system architecture, meaning that it will be possible to change AI models and UI components independently. Code will be well documented so as to help in its maintenance as well as improvement in future.

- **Portability**

The system will be compatible with the popular web browsers such as Firefox, Edge and Chrome. Desktop and mobile users will be accommodated in the system with the assistance of the responsive design.

- **Interoperability**

The system will be integrated into the secure payment gateways such as JazzCash, EasyPaisa, Visa Card, PayPak and Sadapay.

- **Flexibility**

New clothing (e.g. pants, ethnic clothing, kid clothing) will be launched, although without changing core architecture.

5.5 Business Rules

Because there are rules that are expected to regulate the functionality of the system, user permissions and business workflow:

- **User Account Requirement**

Virtual try-on and checkout are only possible to those users who are registered and verified.

- **Admin Privileges**

Only an Admin has the ability to add, edit or delete products in the catalog. Admin has the right to censor uploaded pictures in case misuse is identified.

- **Order Placement Policy**

Orders can be confirmed only after payment has been successfully verified. Cart items are not held awaiting the checkout process.

- **Image Usage Agreement**

Photos that are uploaded by the user should be in accordance with ethical and legal standards (no obscene material). The images are only generated during try on and are not saved unless the user clicks on save result.

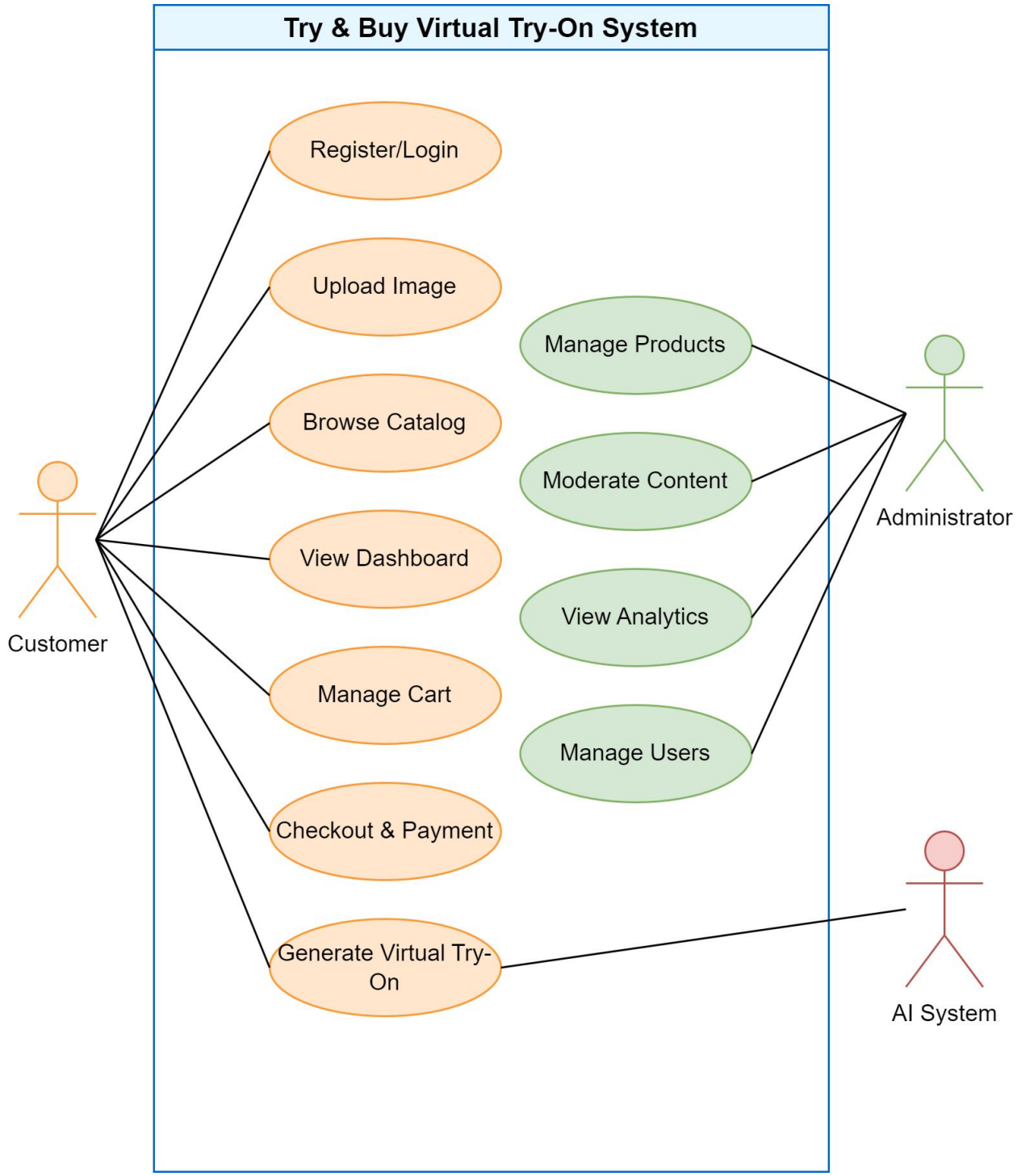
- **Return & Exchange Decision Support**

Virtual try on preview is also given to minimize returns but it is up to the user to make final purchase.

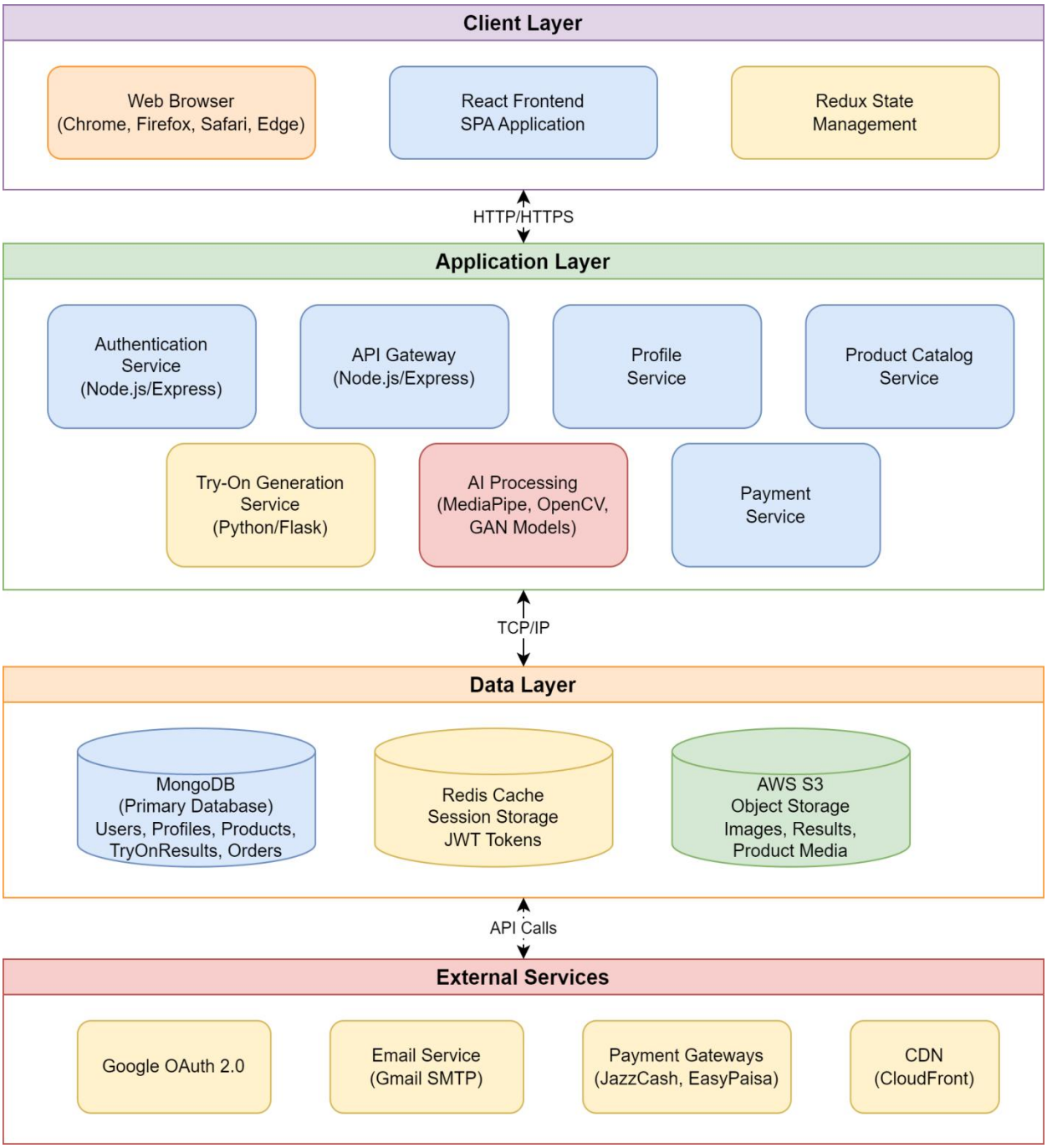
- **Data Privacy Protection**

System shall not provide the personal data to third parties without the user consent. Users can close down their accounts and all the information associated with their names.

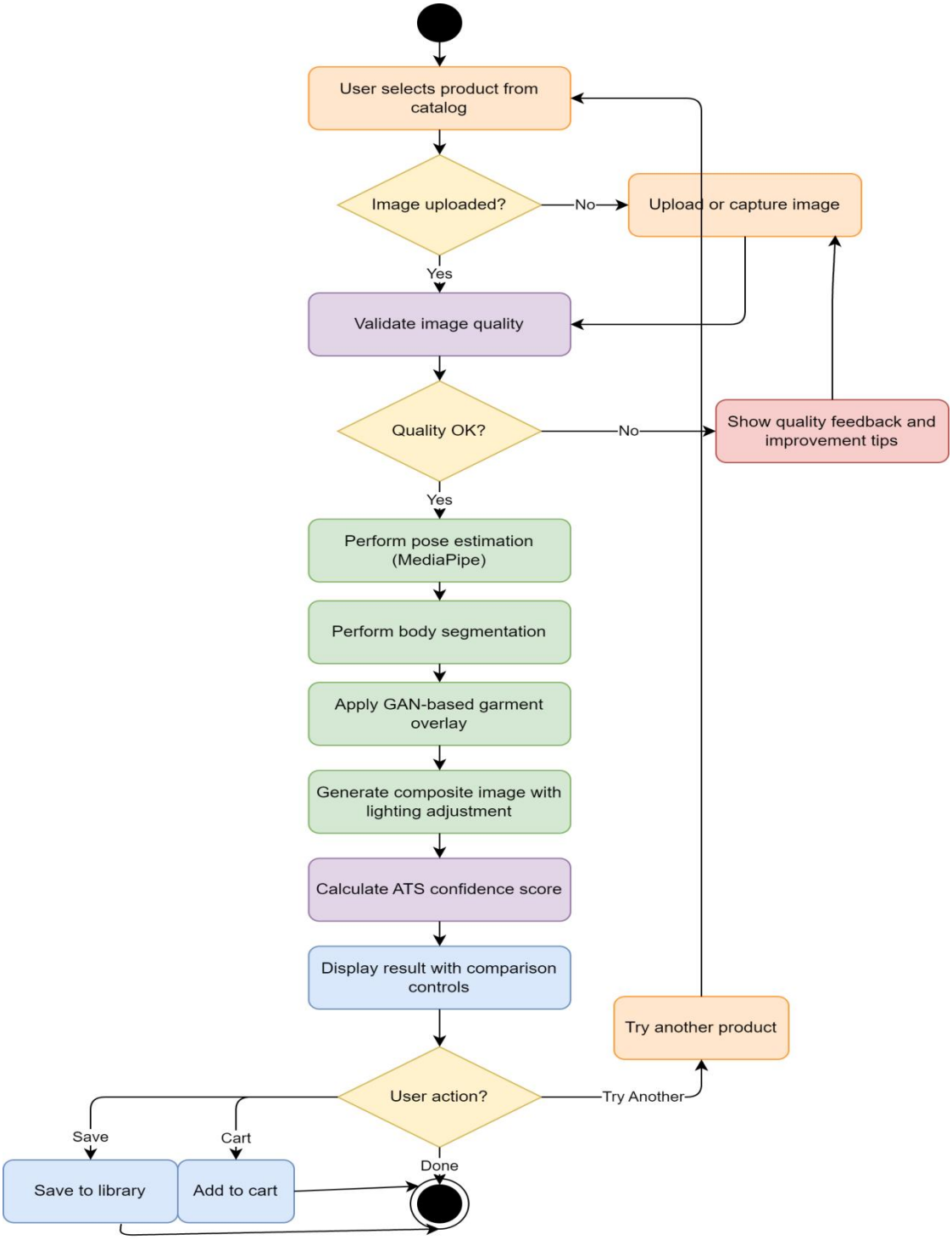
5.6 Use Case Diagram



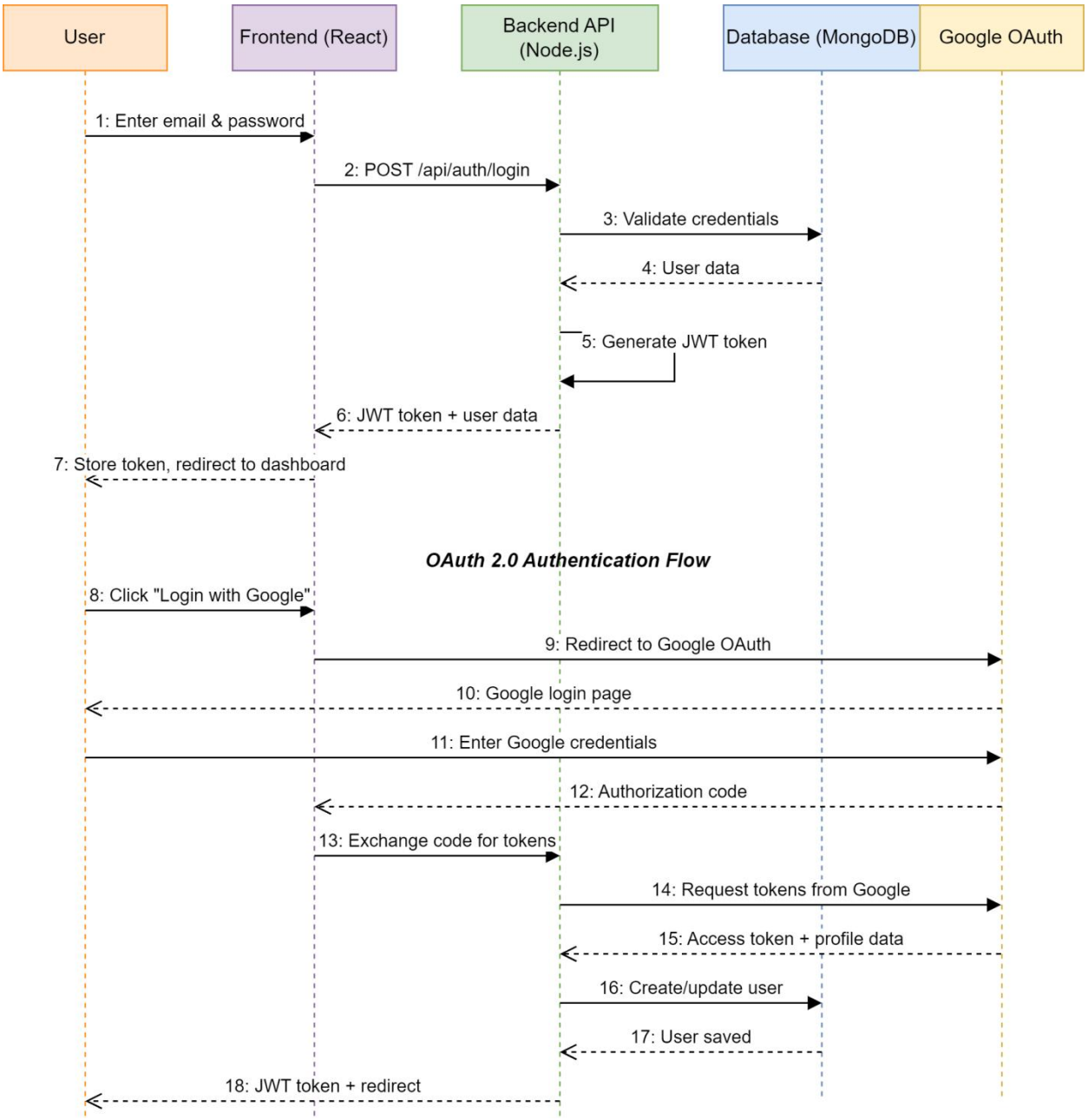
5.7 Architecture Diagram



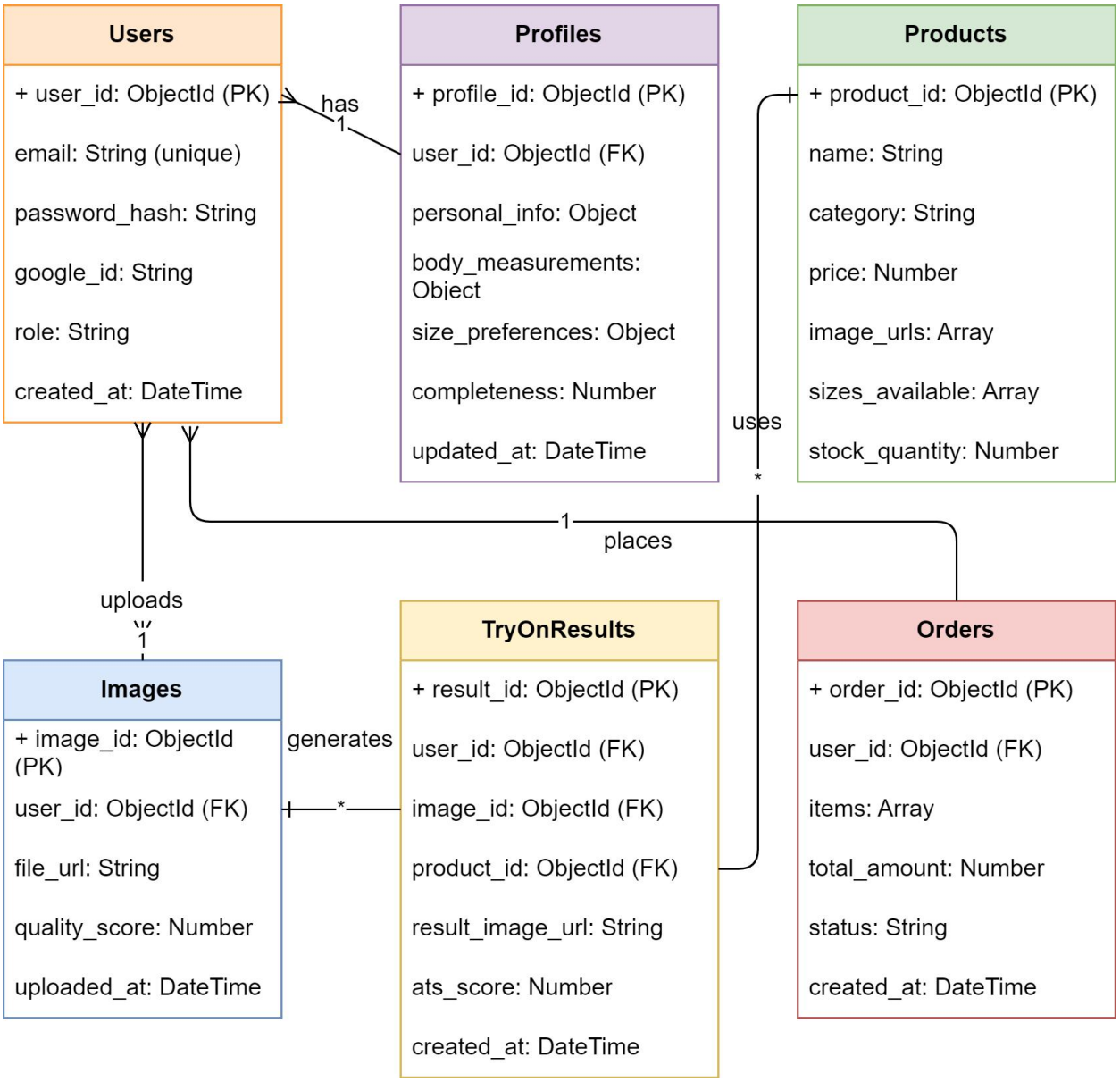
5.8 Activity Diagram



5.9 Sequence Diagram



5.10 ER Diagram



References

[1] Samy, T.M., Asham, B.I., Slim, S.O. et al. Revolutionizing online shopping with FITMI: a realistic virtual try-on solution. Neural Comput & Applic 37, 6125–6144 (2025). Available at: https://link.springer.com/article/10.1007/s00521-024-10843-6

[2] Mediapipe landmark Detector Available at: https://ai.google.dev/edge/mediapipe/solutions/guide

[3] Bradski, G. (2000) The OpenCV Library. Dr. Dobb's Journal of Software Tools, 120; 122-125.

[4] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 8th. McGraw-Hill Education, 2014.

[5] I. Sommerville, Software Engineering, 10th. Pearson, 2015.

[6] https://expressjs.com/

[7] https://www.mongodb.com/

[8] https://react.dev/

[9] https://platform.openai.com/docs/overview

[10] Chen, H., Feng, R., Wu, S. et al. 2D Human pose estimation: a survey. Multimedia Systems 29, 3115–3138 (2023). https://doi.org/10.1007/s00530-022-01019-0

[11] https://medium.com/@codetrade/implementation-of-human-pose-estimation-using-mediapipe-23a57968356b

[12] https://www.ibm.com/think/topics/generative-adversarial-networks

[13] https://www.tensorflow.org/

[14] https://flask.palletsprojects.com/en/stable/